

BACHELOR OF DATA SCIENCE

MAT 201

Linear Algebra

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Reviewed by:

Programme Title	Bachelor of Data Science
Course Title	MAT 201 - Linear Algebra
Learning Module number	Module 1
Learning module title	Review of Linear Algebra and Geometry:
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Reviewed by	

1. COURSE LEARNING GUIDE

1.1. Preparing for Online and Distance Learning

This course offers you the opportunity to enter an online space where you can engage with fellow students at the university. The online forums, exercises, tasks and materials have been designed by the course facilitators in such a way that you can draw directly from their knowledge and share in their scholarly interest in online learning and assessment

We hope you will use your access to this virtual classroom to raise all your questions on the course module, to build a network great learners and to share your own thoughts and experiences to the benefit of your fellow students.

There are a number of steps you can take to ensure you are well prepared to make the most of this learning opportunity. This resource should prompt you to consider the following:

- ☐ How can you best prepare the space where you expect you will spend most of your time engaging in online learning (i.e. in your home or office, or even during your commute to work or whilst traveling)?
- ☐ What can you do to ensure you keep working through the course material, even when you do not have Internet access?
- ☐ How can you familiarise yourself with the course layout, in order to be able to navigate through the steps with ease?
- ☐ What are the communication strategies and best practices you could consider to best engage with your fellow students and the course facilitators?
- ☐ What do you do when you need academic or technical support?

1.2. General Module Guide

The authors of this module believes that imparting knowledge should always be a mutual relationship between the students and the teachers. Students can actively construct knowledge through experiences gained in reading and answering the learning modules.

You should take time to read everything written in the module, actively and religiously solve all the exercises included to finish this course and be able to gain all the competencies and skills expected to demonstrate the learning outcomes as stated in this course. If you finish with full confidence that you understand and can apply all the knowledge you gain in this course, then you are sure to succeed in tackling the next higher subjects in your chosen course.

In this course, you will be taught not only the knowledge you need to demonstrate your learning outcomes, but you will also gain discipline, perseverance, higher thinking skills, and resourcefulness that would help strengthen your foundation to succeed in life. To successfully finish the course, you must follow the guides set forth in using this module.

STUDY ENVIRONMENT.

Before engaging into the module, make sure that you are comfortable enough and diminish distractions. Make sure that you develop a study routine to perform and accomplish all the activities set forth by the module.

MODE OF DELIVERY.

Students are required to get a access a digitised copy of the module from the Virtue Learning Environment. As an open and distance learner, you should be able to learn independently and optimise the learning modes and environment available to you as much of the delivery of all the lessons will be done asynchronously. Your success will greatly depend on your self-motivation and discipline to get your work done without your instructor to remind you from time to time.

STUDY SCHEDULE.

Remember that you have other subjects to attend to, hence, you must manage your time so that you will not be left behind in any of them. Create your own learning plan for each course to make sure that you can perform all the activities. Do not procrastinate your work, remember that time wasted can never be retrieved back. Make sure that even if you are at home, you are a student enrolled in a program, where the learning of all its courses greatly depends on your will to learn as the materials you need are all in your hands. So always be aware of your daily tasks.

PRACTICE EXERCISE.

To measure your learning from your reading, you are given practice exercises which you are supposed to answer daily. This will prepare you in performing your main task.

ONLINE GROUP CHAT.

I am going to create an online Discussion Forum and QA Forum on the VLE. This will serve as a room for discussion on the topics and activities found in your module.

RECOMMENDED REFERENCES FOR SUPPLEMENTARY READING.

You are given links that you need to access online. These links are online references such as e-books, online activities, and tutorial videos of the topic. These are other references you can use to further understand the topic and gain additional knowledge and skills.

MAIN TASK.

You are given a task that will require you to apply the knowledge and skills you learned from the module. This can be a group or individual activity which will measure your higher order thinking skills you acquired throughout the topic.

RUBRICS.

This is a tool to guide you in performing your main task. Through this, you will be able to know the specific indicators that should be found in your task. This is to assess your constructive behaviour towards your learning application. You will be rated according to a 4-point scale shown below.

4 – Excellent 3 –Good 2 – Fair 1 –Poor

REFLECTIONS.

Writing your reflection will be a passageway to express your experiential learning in using this module. In your honest recall, write everything that you think happened to you during your learning experience in this module. Use the guide questions below:

First paragraph.

How was your learning experience? Maybe you can consider describing the effect of the parts of the module (sequence, discussions, examples, practice exercises, online activities, and the main task) in your learning experience. You can point out the strength and weakness of each part so that we can consider it for future improvement of this instructional material.

Second paragraph.

What did you learn? Did you successfully acquire the learning outcomes?

Third paragraph.

How can you use what you learned in the future?

2. ASSESSMENT TASKS & EXAMINATION RULES AND GUIDELINES

1. SUBMISSION OF ASSESSMENT TASKS.

Since mathematical skills are best acquired through constant practice, you are encouraged to answer at least two (2) exercises a day. These practice exercises will not be graded but it will help you develop your thinking skills which will prepare you in the performance of your main tasks. You are required to submit your practice exercises every week.

Usually submission deadlines are set on Saturday 11:59PM EAT

2. GROUP CHAT.

You can post questions, solutions, and comment to other's post. Please be reminded that our group chat is exclusive only for discussion on topics related to this course. Everyone is required to actively participate in the discussions. Also, trolling is highly discouraged. Always observe proper online etiquettes when participating in discussions.

RULE OF THUMB: ALWAYS THINK BEFORE YOU CLICK

3. MAIN TASK.

Complete documentation during the preparation of your task is required to ensure that every member of the group participated and contributed in the task. May it be photos or screenshots of your virtual meetings or group works. You are going to submit it as attachment of your main task the same way as the practice exercises.

4. RECOMMENDED REFERENCES FOR SUPPLEMENTARY READING.

You can explore online for other references that is related to the topic to gain additional information about the topic. Do not share or post inappropriate materials online or even privately.

2.1. REQUIREMENTS TO PASS THE COURSE

Compiled Continuous Assignment/Activities and Tasks with Final Performance Examination.

Grading System

Continuous Activities and Tasks	50%
Final Performance Examination.	50%

2.2. INTRODUCTORY MESSAGE

A warm welcome to **MAT 201** - Linear Algebra, **Module on Review of Linear Algebra and Geometry**.

By this module learning resource, we aim to enable you achieve the relevant competencies, depict skill, action and purpose successfully.

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning material while being an active learner. Your academic success lies in your own hands. This module has the following parts and corresponding icons:

Table 1:

	AUDIENCE	This is a description of a target audience.
	MODULE DESCRIPTION	This part explains the importance of the topics discussed in the module. It will also give you a sneak peek of what you will learn and what is expected from you at the end of the module.
	EXPECTATION	These are what you will be able to know after completing the lessons in the module
	PRE-ACTIVITY	This activity is to get you set, activate and engage schema to learn the module. Through this activity, you will know the relevance of this activity to the topic that will be discussed in the module.
	PRE - ASSESSMENT	This will measure your prior knowledge and the concepts to be mastered throughout the lesson.
	RECAP	This section will measure what learnings and skills that you understand from the previous lesson.
	LESSON	This section will discuss the topic for this module.

Continued on next page

Table 1: (Continued)

	ACTIVITIES	This is a set of activities you will perform.
 Summary	WRAP UP	This section summarizes the concepts and applications of the lessons.
	DISCUSSIONS	This is the meat of the module. The discussion is aligned to the topic learning outcomes that is expected of you to acquire at the end of the module.
	VALUING	This part will check the integration of values in the learning competency you will take home.
	POST - ASSESSMENT	This will measure how much you have learned from the entire module.
	REFLECTION	This part is for you to share your thoughts by writing your reflection, make sure to check your grammar and spelling to ensure the readability and so that it will not alter the thought being delivered.

Audience Description



This module is offered to all students taking the Bachelor of Data Science. As an open and distance learner, you should be able to learn independently and optimise the learning modes and environment available to you. Before you begin this course, please confirm the course material, the course requirements and how the course is conducted.



Module Description

This module provides a comprehensive review of key concepts in linear algebra and geometry, essential for students and professionals in fields such as mathematics, physics, engineering, computer science, and economics. The module covers foundational topics including vectors, dot products, distances, projections, and matrix algebra. It is designed to reinforce understanding and application of these principles in various contexts.



Expectations

This module seeks to enable learning about:

Module Learning Outcomes

By the end of this module, you should be able to:

1. perform basic vector operations such as addition, subtraction, and scalar multiplication.
2. calculate the dot product of two vectors.
3. apply the dot product to solve problems involving orthogonality and vector projections.
4. perform matrix operations including addition, subtraction, scalar multiplication, and matrix multiplication.

3. INTRODUCTION

Question: Why study Linear Algebra?

The study of linear algebra is indispensable for a prospective student of pure, statistics or applied mathematics. It has great value for any undergraduate student who wishes to go beyond the routine manipulations of formulas to solve standard problems, because it develops the ability to think deductively, analyze mathematical situations, and extend ideas to a new context. The subject of linear algebra is one of the fundamental areas of mathematics, and is the foundation for the study of many advanced topics, not only in mathematics, but also in engineering and the physical sciences. A thorough understanding of the concepts of linear algebra has also become increasingly important for the study of advanced topics in economics and the social sciences. Topics such as systems of linear equations, vector spaces and linear transformations are fundamental in mathematics and physics as well as engineering, economics, and many other areas.

Hello and welcome, you are now at the first part of this module. Your first task is to freshen up on the concepts learned from the previous lessons and what you might have learned in high school about vectors and matrices. These tasks are for warm up and will not count towards your final assessment. They are however useful to help you recall concepts that are going to be important in this module.



PRE ASSESSMENT

A complex number z was presented in a vector form as $z = (x, y)$. Let $z = (3, -2)$ and $w = (-4, 6)$ be two complex numbers presented as vectors. Evaluate the magnitudes of z and w , The vector that is the sum $z + w$ and the difference $z - w$, The vector $3z + 7w$, The distance between z and w , What can you say about the relationship between $|z + w|$ and $|z| + |w|$?



RECAP

In MAT 101: Foundations of Mathematics you learned that a complex number can be represented as a vector. That is, a quantity with both magnitude and direction. Vectors play a very important role, and matrices are an example of functions between sets of vectors. In this module, we will formally introduce the concept of vectors, operations on vectors solve problems involving vectors. It will be important to write these vectors in matrix form to make calculations easier.

In most aspects of life, you will encounter linear equations. In most applications, these equations can be rather large and will need to be converted to matrices so that they can be solved. You will be introduced to matrices, albeit smaller ones, and learn basic numerical and algebraic operations on matrices. Some of the most interesting applications of matrices will be learned at a later stage, but here you will be introduced to basic operations. The most important, and widely applied property of matrices that you will learn in this module is the determinant of a square matrix.

You might have learnt of vectors as quantities with magnitude and direction. In this section, you are going to learn the formal definition of a vector, operations on vectors and study some examples highlighting the properties.





Self Pre-recorded Video 1

A video following the lesson on vectors: their definition and operations on vectors.

Vectors

Definition 3.1. A vector is a quantity that has both magnitude and direction. In data analysis, data is frequently arranged in arrays, that is, sets whose elements are indexed. If the data consists of numbers, then a one-dimensional array is called a **vector** i.e., $(a_1, a_2, a_3, \dots, a_n)$ and a two or higher-dimensional array is called a **matrix**. In this section, you will learn about algebraic operations on vectors.

A vector $\mathbf{v}$ in n -dimensional space, $\mathbb{R}^n$, has n components and may be represented as

$$\mathbf{v} = (v_1, v_2, \dots, v_n).$$

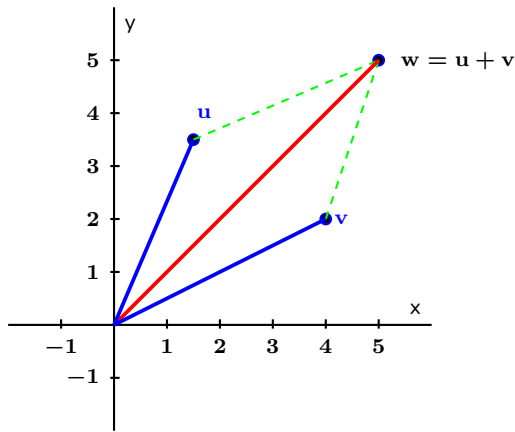
You have already been introduced to complex numbers as vectors $z = (x, y)$ which is in a 2-dimensional space. In this lesson, we will consider the 2-dimensional space and the 3-dimensional space.

Properties of vectors

These operations can only be performed on vectors of the same dimension. Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{R}^n$ and $a, b \in \mathbb{R}$ be scalars. Then

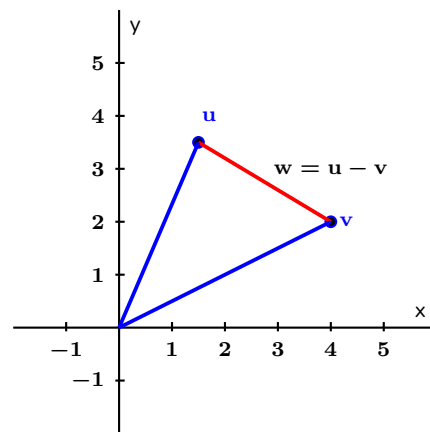
Commutativity	$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
Associativity	$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
Additive inverse	$(-\mathbf{v}) + \mathbf{v} = \mathbf{0} = (0, 0, \dots, 0)$
Distributive	$(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$
Distributive	$a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
Distributive	$(ab)\mathbf{v} = a(b\mathbf{v})$
Multiplicative identity	$1 \cdot \mathbf{v} = \mathbf{v}$

Vectors can be presented geometrically on the cartesian plane as shown below

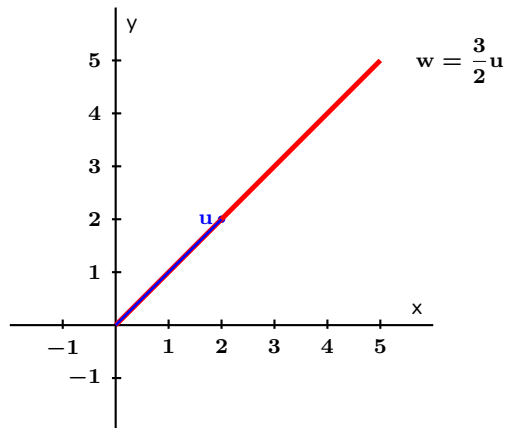


This Shows the sum of two vectors $\mathbf{u}$ and $\mathbf{v}$ to give $\mathbf{w} = \mathbf{u} + \mathbf{v}$

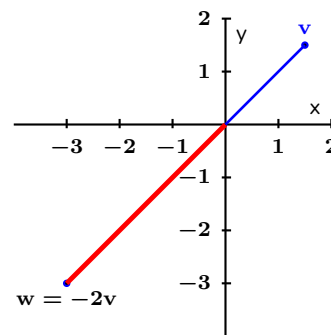
Multiplication of a vector by a non-zero scalar has the effect of either increasing the magnitude, decreasing the magnitude or reversing the direction if the scalar is a negative number. See the graphs below.



This Shows the difference of two vectors $\mathbf{u}$ and $\mathbf{v}$ to give $\mathbf{w} = \mathbf{u} - \mathbf{v}$



This Shows the product of a vector $\mathbf{u}$ by a scalar $\frac{3}{2}$ to give $\mathbf{w} = \frac{3}{2}\mathbf{u}$



This Shows the product of a vector $\mathbf{v}$ by a scalar -2 to give $\mathbf{w} = -2\mathbf{v}$



Example 1:

Let $\mathbf{u} = (-2, 3)$, $\mathbf{v} = (7, 5)$ and $\mathbf{w} = (4, 1)$. Calculate $\mathbf{u} + \mathbf{v} + \mathbf{w}$, $3\mathbf{u} - \mathbf{v} + 2\mathbf{w}$, $4\mathbf{w} + 2\mathbf{u}$.

 Solution

$$\mathbf{u} + \mathbf{v} + \mathbf{w} = (-2, 3) + (7, 5) + (4, 1) = (9, 9)$$

Similarly

$$3\mathbf{u} - \mathbf{v} + 2\mathbf{w} = 3(-2, 3) - (7, 5) + 2(4, 1) = (-5, 6),$$

and

$$4\mathbf{w} + 2\mathbf{u} = 4(4, 1) + 2(-2, 3) = (12, 10).$$

**Example 2:**

Let $\mathbf{u} = (5, -2, 3)$, $\mathbf{v} = (7, 5, 6)$ and $\mathbf{w} = (-1, 0, 1)$. Calculate $\mathbf{u} + \mathbf{v} + \mathbf{w}$, $3\mathbf{u} - \mathbf{v} + 2\mathbf{w}$, $4\mathbf{w} + 2\mathbf{u}$

 Solution

$$\mathbf{u} + \mathbf{v} + \mathbf{w} = (5, -2, 3) + (7, 5, 6) + (-1, 0, 1) = (11, 3, 10)$$

Similarly

$$3\mathbf{u} - \mathbf{v} + 2\mathbf{w} = 3(5, -2, 3) - (7, 5, 6) + 2(-1, 0, 1) = (-1, -16, -1),$$

and

$$4\mathbf{w} + 2\mathbf{u} = 4(-1, 0, 1) + 2(5, -2, 3) = (6, -4, 10).$$

Definition 3.2 (Magnitude of a vector). *The magnitude of a vector $\mathbf{v} = (a_1, a_2, \dots, a_n)$ is given by*

$$|\mathbf{v}| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}.$$

**Example 3:**

Let $\mathbf{u} = (-2, 3)$, $\mathbf{v} = (7, 5, 6)$ and $\mathbf{w} = (1, 3, 1)$. Calculate the magnitude of $\mathbf{u}$, $\mathbf{v} + \mathbf{w}$ and $-\mathbf{v} + 2\mathbf{w}$.

 Solution

$$|\mathbf{u}| = \sqrt{(-2)^2 + 3^2} = \sqrt{13}$$

$$\mathbf{v} + \mathbf{w} = (7, 5, 6) + (1, 3, 1) = (8, 8, 7)$$

so that

$$|\mathbf{v} + \mathbf{w}| = \sqrt{64 + 64 + 49} = \sqrt{177}.$$

Lastly,

$$-\mathbf{v} + 2\mathbf{w} = -(7, 5, 6) + 2(1, 3, 1) = (-5, 1, -4),$$

so that

$$|-\mathbf{v} + 2\mathbf{w}| = \sqrt{(-5)^2 + 1 + (-4)^2} = \sqrt{42}.$$

1. Let $\mathbf{u} = (-3, 5, 1)$, $\mathbf{v} = (0, 2, 1)$ and $\mathbf{w} = (-1, 2, 1)$. Calculate

(a) $\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}$

(c) $|\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}|$

(b) $4\mathbf{u} - \mathbf{u}$

(d) $4|\mathbf{u}| - |\mathbf{u}|$

2. Let $\mathbf{u} = (-3, 3)$, $\mathbf{v} = (2, -3)$ and $\mathbf{w} = (-1, 2)$. Calculate

(a) $\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}$

(c) $|\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}|$

(b) $4\mathbf{u} - \mathbf{u}$

(d) $4|\mathbf{u}| - |\mathbf{u}|$

Definition 3.3 (Column Vectors). When vectors are written vertically instead of horizontally, then they are called column vectors. An example is

$$\mathbf{v} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}.$$

Operations of addition and scalar multiplication can be performed on column vectors just as was the case with row vectors.

Matrices play a very important role in data analysis. Huge data sets are stored in the form of matrices for easy analysis. In this lesson, you will be introduced to matrices, their definition, operation of addition, linear combinations and multiplication of matrices. Pay special attention to multiplication of matrices!





Self Pre-recorded Video 1

A video following the lesson on matrices, the dimension of matrices linear operations on matrices and their properties and multiplication of matrices.

Matrices

Definition 3.4. A matrix A is an $m \times n$ rectangular array of numbers usually presented in the form

$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Expressed in this form, the matrix has m rows and n columns. We call this matrix an $m \times n$ matrix. A matrix with only one row is called a **row matrix** or **row vector**, and a matrix with only one column is called a **column matrix** or **column vector**. A matrix whose entries are all zero is called a **zero matrix** and will usually be denoted by 0 .



Example 4:

Here are some examples of matrices.

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ -1 & 4 \\ -2 & 5 \end{bmatrix}, C = \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \\ 1 & 7 & 6 \end{bmatrix}, D = \begin{bmatrix} 3 & -1 \\ 4 & -2 \end{bmatrix}$$

A is a 2×3 matrix, B a 3×2 matrix, C a 3×3 matrix and D a 2×2 matrix. C and D are examples of square matrices.

Matrix operations

Matrices of the same dimension behave exactly like vectors when performing operations of addition, subtraction and scalar multiplication. Properties of vectors discussed in the previous section apply as well.



Example 5:

Given that $A = \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -1 & 4 \\ -1 & 4 & -2 \end{bmatrix}$, evaluate $A+B$ and $2A-3B$.

Solution

$$\begin{aligned}A + B &= \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix} + \begin{bmatrix} 2 & -1 & 4 \\ -1 & 4 & -2 \end{bmatrix} \\&= \begin{bmatrix} 1+2 & 3+(-1) & -1+4 \\ 4+(-1) & -2+4 & 5+(-2) \end{bmatrix} \\&= \begin{bmatrix} 3 & 2 & 3 \\ 3 & 2 & 3 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}2A - 3B &= 2 \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix} - 3 \begin{bmatrix} 2 & -1 & 4 \\ -1 & 4 & -2 \end{bmatrix} \\&= \begin{bmatrix} 2+(-6) & 6+(3) & -2+(-12) \\ 8+(3) & -4+(-12) & 10+(6) \end{bmatrix} \\&= \begin{bmatrix} -4 & 9 & -14 \\ 11 & -16 & 16 \end{bmatrix}\end{aligned}$$



Example 6:

Given that $M = \begin{bmatrix} 3 & -1 \\ 2 & -1 \\ 4 & -2 \end{bmatrix}$, $N = \begin{bmatrix} 5 & -1 \\ 2 & 4 \\ 4 & -2 \end{bmatrix}$, evaluate $2N - M$ and $-8N$.

Solution

$$\begin{aligned}2N - M &= 2 \begin{bmatrix} 5 & -1 \\ 2 & 4 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 3 & -1 \\ 2 & -1 \\ 4 & -2 \end{bmatrix} \\&= \begin{bmatrix} 10 - 3 & -2 - 1 \\ 4 - 2 & 8 - (-1) \\ 8 - 4 & -4 - (-2) \end{bmatrix} \\&= \begin{bmatrix} 7 & -3 \\ 2 & 9 \\ 4 & 2 \end{bmatrix}\end{aligned}$$

and

$$\begin{aligned}-8N &= -8 \begin{bmatrix} 5 & -1 \\ 2 & 4 \\ 4 & -2 \end{bmatrix} \\&= \begin{bmatrix} -40 & 8 \\ -16 & -32 \\ -32 & 16 \end{bmatrix}\end{aligned}$$

1. Let $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 3 & 0 \\ 4 & 4 & 1 \end{bmatrix}$, evaluate $2A + 3B$ and $4A - B$.
2. Given that $M = \begin{bmatrix} 1 & 0 & 5 \\ -1 & 3 & 2 \\ 3 & 1 & -1 \end{bmatrix}$, $N = \begin{bmatrix} 2 & 2 & -1 \\ 4 & -1 & 2 \\ 0 & 4 & 0 \end{bmatrix}$, evaluate $3N + N$ and $5M - 2N$.

Matrix Multiplication

Definition 3.5. Given a row matrix $A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$ and a column matrix $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$,

the product AB is defined as

$$AB = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = a_1b_1 + a_2b_2 + \cdots + a_nb_n.$$

This is not a matrix but a number.

In general, to multiply two matrices A and B, the number of columns of matrix A must be equal to the number of rows of matrix B for the multiplication to make sense. To get the product AB, rows of A are multiplied with the columns of B to get the entries of the matrix AB. See the example below.



Example 7:

Let $A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}$. Evaluate the products AB and BA.

Solution

The dimensions of B are 3×2 and the dimensions of A are 2×3 . The inner dimensions match so the product is defined and will be a 3×3 matrix.

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 1 & -1 \\ 13 & 5 & -1 \\ 1 & 0 & -2 \end{bmatrix}. \end{aligned}$$

On the other hand, the dimensions of A are 2×3 and the dimensions of B are 3×2 . The inner dimensions match so the product is defined and will be a 2×2 matrix.

$$\begin{aligned} BA &= \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 3 \\ 6 & 1 \end{bmatrix} \end{aligned}$$

Notice that AB and BA are not equal! This illustrates the fact that matrix multiplication is NOT commutative.

Definition 3.6 (Properties of Matrix Multiplication). *For matrices A, B and C, the following properties hold.*

1. *Matrix multiplication is Associative:* $(AB)C = A(BC)$.
2. *Matrix multiplication is distributive*

$$\begin{aligned} A(B + C) &= AB + AC \\ (A + B)C &= AC + BC \end{aligned}$$

As we have observed from the previous example. matrix multiplication is NOT commutative.

1. Given the matrices $A = \begin{bmatrix} 2 & 3 \\ 3 & -2 \\ -4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 6 \\ -3 & 7 \\ 6 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & -1 & 3 \\ -3 & -2 & 2 \end{bmatrix}$ Evaluate
- AC and CA ,
 - $(A + B)C$ and $C(A + B)$
 - BC and CB
 - Verify that $(A + B)C = AC + BC$ and $C(A + B) = CA + CB$
2. Given the square matrices A , B and C below, evaluate AB , BA , $B(A + C)$ and $A(B + C)$ and $(A + B)C$
- $A = \begin{bmatrix} 6 & 6 \\ 3 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 7 \\ 10 & -2 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$.
 - $A = \begin{bmatrix} 8 & 5 & 2 \\ 2 & 7 & 2 \\ 5 & 6 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 10 & 6 \\ 5 & -1 & 7 \\ 2 & 2 & 10 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 6 & 10 \\ -3 & 0 & 10 \\ -1 & 0 & -5 \end{bmatrix}$.

Square matrices are very special. Each square matrix can be evaluated for the determinant, which tells us if the matrix has an inverse or not. In this section, you will learn about various ways of calculating the determinant of a square matrix, and properties of determinants.



Self Pre-recorded Video 1

A video following the lesson on calculation of determinants. Use examples to explain calculation of the determinant using row operations, and using the cofactor method.

The determinant of a square matrix

Definition 3.7. Let A be a square $n \times n$ matrix. The determinant of A denoted by $|A|$ or $\det(A)$ is a real number and contains important information about the matrix. We can think of the **determinant as a function**

$$\det : \{\text{Set of square matrices}\} \rightarrow \mathbb{R}$$

The determinant satisfies the following properties

- Doing a row replacement on A (that is a linear operation on the rows of A) does not change $\det(A)$.
- Scaling a row of A (i.e. multiplying a row by a non-zero constant) by a scalar c multiplies the determinant by c .
- Swapping two rows of a matrix multiplies the determinant by -1 .
- The determinant of the identity matrix I_n is equal to 1.
- If the determinant of A is non-zero, then A has an inverse A^{-1} and $\det(A^{-1}) = \frac{1}{\det(A)}$.
- If A and B are two matrices of the same dimension, then $\det(AB) = \det(A)\det(B)$.

There are several ways of evaluating the determinant of square matrices. It is important to know how this is done by hand, but in real life situations, the matrices can be very large, and so calculations are done using Computer Algebra Engines like python, sage, mathematica, matlab, maxima e.t.c.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a non zero matrix. Then

$$\det(A) = ab - cd.$$

For ways of evaluating determinants of higher dimensional matrices, see the reading activity below.

Activity 1: Reading activity

Read [4, Section 4] on definition of the determinant and the calculations of determinants using row operations and using the cofactor method. Then attempt the problems below.



Example 8:

Let $A = \begin{bmatrix} 5 & 6 \\ 3 & -2 \end{bmatrix}$. Evaluate the determinant of A.

Solution

$$\det(A) = 5(-2) - 3(6) = -28.$$



Example 9:

Let $A = \begin{bmatrix} 3 & -2 \\ 5 & 6 \end{bmatrix}$. Evaluate the determinant of A.

Solution

$$\det(A) = 3(6) - 5(-2) = 28.$$

From the two examples above, we can see that interchanging the rows of A has the effect of changing the sign of the determinant.



Example 10:

Use the cofactor method to evaluate the determinant $\det(C)$ of $C = \begin{bmatrix} 2 & 6 & 5 \\ 3 & 7 & -1 \\ 4 & 0 & 9 \end{bmatrix}$



Solution

Using the cofactor method,

$$\begin{aligned} \begin{vmatrix} 2 & 6 & 5 \\ 3 & 7 & -1 \\ 4 & 0 & 9 \end{vmatrix} &= 2 \begin{vmatrix} 7 & -1 \\ 0 & 9 \end{vmatrix} - 6 \begin{vmatrix} 3 & -1 \\ 4 & 9 \end{vmatrix} + 5 \begin{vmatrix} 3 & 7 \\ 4 & 0 \end{vmatrix} \\ &= 2(63 - 0) - 6(27 - (-4)) + 5(0 - 28) \\ &= -200 \end{aligned}$$



Example 11:

Use the cofactor method to evaluate the determinant $\det(C)$ of $C = \begin{bmatrix} 2 & 6 & 5 \\ 3 & 7 & -1 \\ 4 & 0 & 9 \end{bmatrix}$



Solution

Using the cofactor method,

$$\begin{aligned} \begin{vmatrix} 2 & -2 & 4 \\ 6 & 1 & 7 \\ -1 & 4 & -1 \end{vmatrix} &= 2 \begin{vmatrix} 1 & 7 \\ 4 & -1 \end{vmatrix} - (-2) \begin{vmatrix} 6 & 7 \\ -1 & -1 \end{vmatrix} + 4 \begin{vmatrix} 6 & 1 \\ -1 & 4 \end{vmatrix} \\ &= 2(-1 - 28) + 2(-6 + 7) + 4(24 + 1) \\ &= 44 \end{aligned}$$

Evaluate the determinants of the following matrices

1. $\begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix},$

2. $\begin{bmatrix} 10 & 0 & -3 \\ 2 & -3 & 7 \\ 2 & 1 & -5 \end{bmatrix},$

3. $\begin{bmatrix} -4 & 4 & 0 \\ 6 & 3 & 2 \\ -5 & -3 & -1 \end{bmatrix}$



If we think of matrices as functions (mappings) between sets, then matrices with non-zero determinants play a key role because they have inverses. Determinants of these matrices can then be used to calculate the area and volume of geometric objects. You will find this application very interesting indeed!



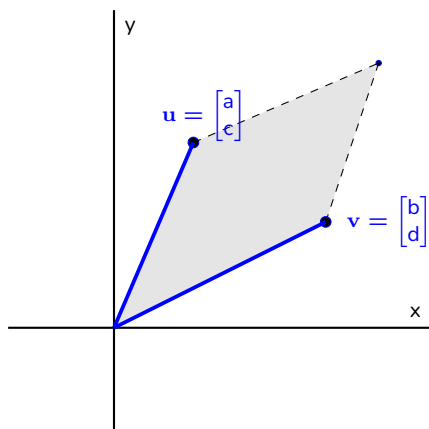
Self Pre-recorded Video 1

A video following the application of determinants in calculating the area of geometric shapes. Of consideration for our context is the area of a parallelogram and the volume of parallelepiped.

Applications of the determinant

Determinants of matrices has very many applications. One of the applications is in solving systems of linear equations, which you will encounter in a later course. We are going to look at the application in evaluating the area and volume of geometric shapes.

Consider the parallelogram in $\mathbb{R}^2$ whose sides are vectors $\mathbf{u} = \begin{bmatrix} a \\ c \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} b \\ d \end{bmatrix}$ in the diagram below



The area of the parallelogram is given as the absolute value of the determinant of the matrix

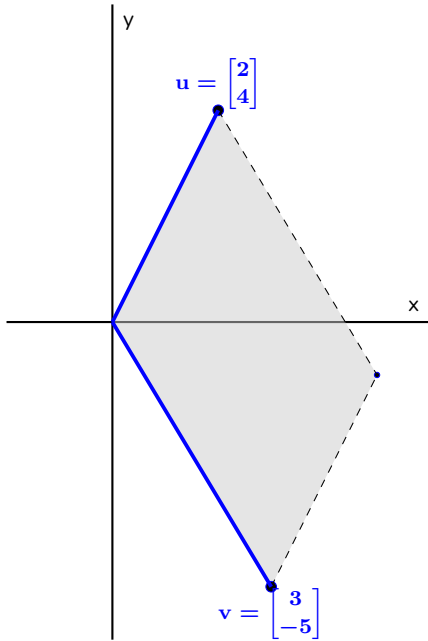
$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. That is

$$\text{Area} = |\det(A)| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = |ad - cb|.$$



Example 12:

Calculate the area of the parallelogram with sides $(2, 4)$ and $(3, -5)$. See the diagram below.



Solution

Here, form the matrix $A = \begin{bmatrix} 2 & 3 \\ 4 & -5 \end{bmatrix}$. The area of the parallelogram will be

$$\text{Area} = |\det(A)| = \begin{vmatrix} 2 & 3 \\ 4 & -5 \end{vmatrix} = |-10 - 12| = 22 \text{ Square units.}$$

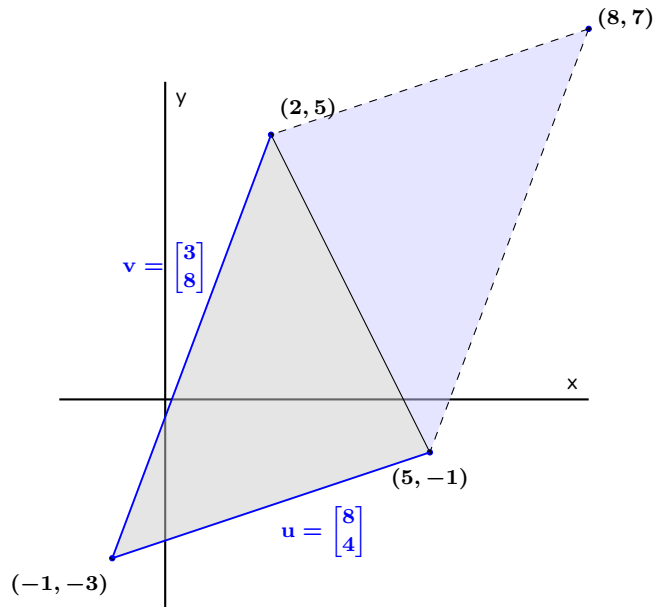


Example 13:

Calculate the area of the triangle with vertices at $(-1, -3)$, $(2, 5)$ and $(5, -1)$.

Solution

This is just one half of a parallelogram with vertices $(-1, -3)$, $(2, 5)$, $(5, -1)$ and $(8, 7)$. See the diagram below.



We express the sides of the parallelogram as vectors $\mathbf{u} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$. The area of the triangle will be given by

$$\text{Area} = \frac{1}{2} |\det(\mathbf{A})| = \frac{1}{2} \begin{vmatrix} 3 & 8 \\ 8 & -1 \end{vmatrix} = \frac{1}{2} |-3 - 40| = 21.5 \text{ Square units.}$$

More applications of determinants in the calculations of areas and volumes can be found in the reading activity below.

Activity 2: Reading activity

Read on [4, Section 4] on the applications of determinants in calculating areas and volumes then attempt the exercise below.

1. Calculate the area of the triangle with vertices at
 - (a) $(-1, -2)$, $(3, 8)$ and $(9, 0)$.
 - (b) $(-5, 5)$, $(5, 4)$ and $(3, -3)$.
2. Calculate the area of a parallelogram whose vertices are given by $(-5, 5)$, $(-4, -3)$, $(3, -2)$ and $(2, 6)$.

To complete this module, take the end of module assessment below. This will contribute to your overall assessment. The test is worth 25 marks



POST ASSESSMENT



1. Open Ended Questions

Solve the following problems to complete your assessment for this module.

- (a) Let $\mathbf{u} = (-3, 5, 1)$, $\mathbf{v} = (0, 2, 1)$ and $\mathbf{w} = (-1, 2, 1)$. Calculate $7\mathbf{u} - 3\mathbf{v} - \mathbf{w}$ and $|\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}|$
- (b) Let $C = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 3 \\ 4 & 4 \end{bmatrix}$, evaluate $2A + 3B$ and $4A - B$.
- (c) Given the matrices $A = \begin{bmatrix} 2 & 3 \\ 3 & -2 \\ -4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 6 \\ -3 & 7 \\ 6 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & -1 & 3 \\ -3 & -2 & 2 \end{bmatrix}$ Evaluate AC and CA , $(A + B)C$ and $C(A + B)$.
- (d) Evaluate the determinants of the following matrices
- i. $\begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix}$,
- ii. $\begin{bmatrix} 10 & 0 & -3 \\ 2 & -3 & 7 \\ 2 & 1 & -5 \end{bmatrix}$,
- (e) Calculate the area of the triangle with vertices at $(3, -3)$, $(-4, -3)$ and $(-1, 2)$.



WRAP UP

Summary

To wrap-up, answer the following questions:

1. What is a vector?
2. What is a matrix?
3. can two matrices be multiplied? What properties do they need to satisfy for the multiplication to make sense?
4. Is the product of two matrices commutative?
5. What is the determinant of a square matrix, and how is it calculated?
6. How can the determinant be used to calculate the area of geometric objects?



VALUING AND TAKE HOME



Hello again, you are now at the last part of this module. Your task is to do a self-reflection on your learning experience.

Vectors are important mathematical objects and so are matrices. You will be encountering matrices and vectors in other courses and in your day to day life. Knowing the definitions and properties helps in the understanding of these mathematical objects. In real life, you will be dealing with very large vectors and matrices of very high dimensions. Some of the calculations done in this course will not be possible by hand, and so you will need to use computer algebra programs to do your calculations. It will be important to have an intuition of what these operations will entail.



DISCUSSION AND REFLECTION

Can you think of how determinants can be used to calculate the area of an ellipse? Give a practical example.



CORE READING AND REFERENCES

References

- [1] Dennis G. Zill and Patrick D. Shanahan. (2015). Complex Analysis: A First Course with Applications 3rd Edition. Jones and Bartlett Learning. ISBN-13 978-1449694616.
- [2] Lary Green. Determinants and Inverses. (2021, March 5). Lake Tahoe Community College. Libre Texts. <https://math.libretexts.org/@go/page/40861>
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- [4] Margalit, D., and Rabinoff, J. (2022, April 3). Determinants. Georgia Institute of Technology. Libre Texts. <https://math.libretexts.org/@go/page/70201>
- [5] Serge Lang, 2000, Basic Mathematics, Adison-Wesley. ISBN 978-1-4612-1027-6

[subsection]Definition [subsection]Corollary [subsection]Proposition [subsection]Theorem [subsection]Lemma [subsection]Remark



Module Description

This module delves into the methods and applications of solving systems of linear equations, a cornerstone topic in linear algebra. Participants will explore matrix operations, Gaussian elimination, determinant calculations, and various techniques for solving linear systems. The module emphasizes both theoretical understanding and practical applications, such as polynomial curve fitting and geometric interpretations.



Expectations

This module seeks to enable learning about:

Module Learning Outcomes

By the end of this module, you should be able to:

1. describe elementary row and column operations, echelon forms, and Gaussian Elimination.
2. use Gaussian elimination, Cramer's and LU factorization to solve linear system.
3. assess the consistency and uniqueness of solutions using determinants and matrix rank.
4. solve real-world problems such as polynomial curve fitting and determining the equation of a plane through three points

4. SOLVING A SYSTEM USING ELEMENTARY OPERATIONS

4.1. Gaussian Elimination

Definition 4.2. Two systems of linear equations are called equivalent if they have the same solution set.



Example 14:

Solve the system

$$\begin{aligned}x - 2y + 3z &= 9 \\y + 3z &= 5 \\z &= 2\end{aligned}$$

Solution

By a method called back-substitution, we find that $z = 2$, and $y = 5 - 3(2) = -1$ and $x = 9 + 2(-1) - 3(2) = 1$.

Definition 4.3. A zero row of a matrix is a row that consists entirely of zeros. A nonzero row is a row that has at least one nonzero entry.

In the same manner, we can talk about zero and nonzero columns. The first nonzero entry of a nonzero row is called a leading entry. If a leading entry happens to be 1, we call it a leading 1. We say that the system is in echelon form, if it follows a stair step pattern and has leading coefficients of 1.

Definition 4.4. An $m \times n$ matrix C is in echelon form if

1. All rows that consist entirely of zeros of zeros are grouped together at the bottom of the matrix.
2. The first (counting from left to right) nonzero entry in the $(i + 1)$ st row must appear in a column to the right of the first nonzero entry in the i th row. (Many definitions of echelon form require that the first nonzero entry in each row be a leading 1, but this

not necessary). This means that for two successive nonzero rows, the leading 1 in the higher row is farther to the left than the leading 1 in the lower row. Equivalently, the first nonzero entry on each row is found to the right of the first nonzero entry in the preceding row.

To solve a system that is not in row-echelon form, we first change it to an equivalent system that is in echelon form by using some elementary operations, which can be applied to a system of linear equations without fundamentally changing the system.

Definition 4.5. The following operations, performed on the rows of a matrix /linear system, are called elementary row operations:

1. Interchange two rows of the matrix.
2. Multiply a row/equation by a nonzero scalar. That is replace a row/equation by a scalar multiple of itself.
3. Add a constant multiple of one row/equation to another. That is, replace one row/equation by adding the scalar multiple of another row/equation.

Remark 4.6. Rewriting a system of linear equations in row-echelon form usually involves a chain of equivalent systems, each of which is obtained by using one of the three basic operations. This process is called Gaussian Elimination (named after the German mathematician Karl Friedrich Gauss, 1777-1855)

Definition 4.7. A matrix is said to be in reduced row echelon form (RREF) if it is already in echelon form and satisfies the following conditions:

3. The leading entry in any nonzero row is 1.
4. All entries in the column above and below a leading 1 are zero.

Note that a reduced row echelon form is always in echelon form.



Example 15:

Let

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}, C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{pmatrix}, E = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, F = \begin{pmatrix} 1 & 7 & 0 & 9 & 0 \\ 0 & 0 & 1 & -8 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$G = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, H = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

Clearly A, B, D, F, G, H are in echelon form, because the conditions (1) and (2) above hold. Of these, A, B, D, F are in RREF, because all the four conditions (1), (2), (3) and (4) above hold. G and H are not in RREF. For G condition (4) fails. For H condition (3) fails. C and E are not in echelon form (and hence cannot be in RREF). For C condition

(2) fails and for E, condition (1) fails.

Definition 4.8. Two matrices are (row) equivalent if one can be obtained from the other by a finite sequence of elementary row operations. We use the abbreviation

$$A \sim B$$

for the statement "matrix A is equivalent to B".

Remark 4.9. We say that a matrix reduces to RREF if it is equivalent to a matrix in RREF. It can be shown that every matrix is row equivalent to a matrix in RREF.

5. Gauss Elimination Algorithm

To reduce any matrix to RREF, apply the following steps:

Step 1: Find/locate the leftmost nonzero column.

Step 2: If the first row has a zero in the column of Step 1, interchange it with one that has a nonzero entry in the same column.

Step 3: Obtain zeros below the leading entry by adding suitable multiples of the top row to the rows below that.

Step 4: Cover the top row and repeat the same process starting with Step 1 applied to the leftover sub-matrix. Repeat this process with the rest of the rows. (At this stage the matrix is already in echelon form).

Step 5: Starting with the last nonzero row, work upward: For each row obtain a leading 1 and introduce zeros above it by adding suitable multiples to the corresponding rows.

Remark 5.1. The first four steps of the algorithm are known as the forward pass of the Gauss elimination. They bring the matrix to echelon form. Step 5 is the backward pass (or back-substitution). This step reduces the matrix to reduced row echelon form (RREF).

- Gauss elimination brings a matrix or system into triangular form. Recall that a matrix $A = [a_{ij}]$ is upper triangular if $a_{ij} = 0$ whenever $i > j$ and lower triangular if $a_{ij} = 0$ whenever $i < j$.
- To solve the system of linear equations $A\bar{x} = \bar{b}$, we apply Gauss elimination or elementary row operations (ERO) to the augmented matrix $[A \mid \bar{b}]$ and try to reduce it to triangular form.



Example 16:

The following system is in upper triangular form

$$\begin{pmatrix} 5 & -1 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 11 \\ 2 \\ 12 \end{pmatrix}$$

with augmented matrix

$$\left(\begin{array}{ccc|c} 5 & -1 & 1 & 11 \\ 0 & 2 & -1 & 2 \\ 0 & 0 & 3 & 12 \end{array} \right)$$

This augmented matrix represents the linear system

$$\begin{aligned} 5x - y + z &= 11 \\ 2y - z &= 2 \\ 3z &= 12 \end{aligned}$$

Solution

A solution for the last variable can be found by inspection. We see that $z = 4$. To find the values of the other values, we use back-substitution. We substitute the value we have found into the equation immediately above it. In this case, insert the value found for z into the second equation. This allows us to solve for y : $2y - z = 2, z = 4 \Rightarrow y = 3$. Now substitute the values $y = 3$ and $z = 4$ into the first equation to obtain $x = 2$.

Once a system has been reduced, we call the coefficient of the first unknown in each row a pivot. For example in the reduced system above, the pivots are 5, 2 and 3.



Example 17:

Using Gaussian elimination find the solution to the system

$$\begin{aligned} 2y - z &= 1 \\ -x + 2y - z &= 0 \\ x - 4y + z &= 2 \end{aligned}$$

Solution

The augmented matrix is

The first pivot position contains a zero. So we interchange R_1 and R_3 (i.e. $R_1 \longleftrightarrow R_3$) to move a nonzero value into the first pivot. This gives

$$\left(\begin{array}{ccc|c} 1 & -4 & 1 & 2 \\ -1 & 2 & -1 & 0 \\ 0 & 2 & -1 & 1 \end{array} \right)$$

There is only one term to eliminate below the first pivot. We add the first row to the second row (i.e. $R_2 := R_2 + R_1$) to get

$$\left(\begin{array}{ccc|c} 1 & -4 & 1 & 2 \\ 0 & -2 & 0 & 2 \\ 0 & 2 & -1 & 1 \end{array} \right)$$

The second row tells us

$$-2y = 2$$

Therefore $y = -1$. We substitute this value into the third equation to get

$$z = 2y - 1 = -2 - 1 = -3$$

We can then find x using the equation in the top row:

$$x = 4y - z + 2 = -4 + 3 + 2 = 1$$

5.2. HOMOGENEOUS SYSTEMS OF LINEAR EQUATIONS

A homogeneous system is a linear system with all zeros on the right hand side. In general, it is a system of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= 0 \end{aligned} \tag{1.2}$$

Remark 5.3. When a homogeneous system is put in echelon form, if the system has more unknowns than it has a nonzero solution. A system in echelon form with n equations and n unknowns has only the zero (trivial) solution, meaning that $(x_1, x_2, \dots, x_n) = (0, 0, \dots, 0)$.



Example 18:

Use Gaussian elimination to determine if the system

$$\begin{aligned} 2x - 8y + z &= 0 \\ x + y - z &= 0 \\ 3x + 3y + 2z &= 0 \end{aligned}$$

has a solution.  **Solution**

We bring the augmented matrix into echelon form using the elementary operations:

$R_2 := R_2 - \frac{1}{2}R_1, R_3 := \frac{1}{3}R_3, R_3 := R_3 - \frac{1}{2}R_1$ and finally $R_3 := R_3 - R_2$ we get

the reduced matrix The augmented matrix is

The system is now in echelon form. There are three equations and three unknowns, and therefore the system has only the zero (trivial) solution $(x, y, z) = (0, 0, 0)$.



Example 19:

Solve the following system of linear equations

$$\begin{aligned}x_1 - x_2 + 3x_3 &= 0 \\ 2x_1 + x_2 + 3x_3 &= 0\end{aligned}$$

Solution

The augmented matrix reduces as follows

$$\left(\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 2 & 1 & 3 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right)$$

after applying the elementary operations (in the given order) $R_2 := R_2 - 2R_1$, $R_2 := \frac{1}{3}R_2$, $R_1 := R_1 + R_2$. Therefore the system of equations correspond to

$$\begin{aligned}x_1 + 2x_3 &= 0 \\ x_2 - x_3 &= 0\end{aligned}$$

Using the parameter $x_3 = t$, we conclude that the solution set is given by $x_1 = -2t$, $x_2 = t$, $x_3 = t$, where t is any real number. Note that the trivial solution (when $t = 0$) is also a solution of this system.

Theorem 1.1 Every homogeneous system of linear equations is consistent. Moreover, it is understood that if the system has fewer equations than variables, then it must have an infinite number of solutions.

6. APPLICATIONS OF SYSTEMS OF LINEAR EQUATIONS

Systems of linear equations arise in a wide variety of applications. One of these is in polynomial curve fitting.

6.1. 1.5.1 Polynomial Curve Fitting

Suppose a collection of data is represented by n points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ in the xy -plane and we are asked to find a polynomial

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^n$$

of degree $n - 1$ whose graph passes through the given points. This procedure is called polynomial curve fitting.

It can be shown that if all the x -coordinates of the points are distinct, then there is precisely one polynomial function of degree $n - 1$ (or less) that fits the n points.

Figure 1.1: Polynomial curve fitting

To solve for the n coefficients $a_0, a_1, \dots, a_{n-1}$ of the polynomial $p(x)$, we substitute each of the n points into the polynomial and obtain n linear equations in the variables $a_0, a_1, \dots, a_{n-1}$:

$$\begin{array}{rcl} a_0 + a_1x_1 + a_2x_1^2 + \dots + a_{n-1}x_1^{n-1} & = & y_1 \\ a_0 + a_1x_2 + a_2x_2^2 + \dots + a_{n-1}x_2^{n-1} & = & y_2 \\ \vdots & & \vdots \\ a_0 + a_1x_n + a_2x_n^2 + \dots + a_{n-1}x_n^{n-1} & = & y_n \end{array}$$



Example 20:

Determine the polynomial $p(x) = a_0 + a_1x + a_2x^2$ whose graph passes through the points $(1, 4)$, $(2, 0)$ and $(3, 12)$.  **Solution**

Substituting $x = 1, 2$ and 3 into $p(x)$ and equating the results to the respective y -values produces the following system of linear equations in the variables a_0, a_1 and a_2 .

$$p(1) = a_0 + a_1(1) + a_2(1)^2 = a_0 + a_1 + a_2 = 4$$

$$p(2) = a_0 + a_1(2) + a_2(2)^2 = a_0 + 2a_1 + 4a_2 = 0$$

$$p(3) = a_0 + a_1(3) + a_2(3)^2 = a_0 + 3a_1 + 9a_2 = 12$$

The solution of this system is $a_0 = 24$, $a_1 = -28$ and $a_2 = 8$. Thus the polynomial is

$$p(x) = 24 - 28x + 8x^2$$



Example 21:

Find a polynomial that fits the points $(-2, 3)$, $(-1, 5)$, $(0, 1)$, $(1, 4)$ and $(2, 10)$.

 **Solution**

Since we are given five points we need to find a fourth degree polynomial

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$$

Substituting the given values of x into $p(x)$ gives the following system

$$\begin{aligned} a_0 - 2a_1 + 4a_2 - 8a_3 + 16a_4 &= 3 \\ a_0 - a_1 + a_2 - a_3 + a_4 &= 5 \\ a_0 &= 1 \\ a_0 + a_1 + a_2 + a_3 + a_4 &= 4 \\ a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 &= 10 \end{aligned}$$

This system has solution

$$a_0 = 1, a_1 = -\frac{30}{24}, a_2 = \frac{101}{24}, a_3 = \frac{18}{24}, a_4 = -\frac{17}{24}$$

Thus the polynomial is given by

$$p(x) = 1 - \frac{30}{24}x + \frac{101}{24}x^2 + \frac{18}{24}x^3 - \frac{17}{24}x^4$$

6.2. 1.5.2 Equation of a Plane Through Three Points



Example 22:

Find the equation of the plane in the xyz -space passing through the points $P(1, 1, 2)$, $Q(1, 2, 0)$ and $R(2, 1, 5)$.  **Solution**

Let $ax + by + cz + d = 0$ be the equation of the plane. We need to determine the coefficients a, b, c and the constant d . Because P belongs to the plane, we should have $a(1) + b(1) + c(2) + d = 0$. Likewise, using the other two points we get the homogeneous system

$$\begin{aligned} a + b + 2c + d &= 0 \\ a + 2b + d &= 0 \\ 2a + b + 5c + d &= 0 \end{aligned}$$

Solving yields the one-parameter infinite solution set $a = 3t, b = -2t, c = -t$ and $d = t$. If we let $d = t = 1$, then we get the plane

$$3x - 2y - z + 1 = 0$$

Any other values of t gives a constant multiple of this equation, which represents the same plane.



ACTIVITIES

1. Find the parametric representation of a solution set of the linear equation $3x + 2y - 2z = 9$
2. Solve the following system by Gaussian elimination.

7. Rank and Homogeneous Systems

Our focus in this section is to consider what types of solutions are possible for a homogeneous system of equations. Consider the following definition. Consider the homogeneous system of equations given by

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0\end{aligned}$$

Then, $x_1 = 0, x_2 = 0, \dots, x_n = 0$ is always a solution to this system. We call this the trivial solution.

If the system has a solution in which not all of the $x_1, \dots, x_n$ are equal to zero, then we call this solution nontrivial. The trivial solution does not tell us much about the system, as it says that $0 = 0$! Therefore, when working with homogeneous systems of equations, we want to know when the system has a nontrivial solution.

Suppose we have a homogeneous system of m equations, using n variables, and suppose that $n > m$. In other words, there are more variables than equations. Then, it turns out that this system always has a nontrivial solution. Not only will the system have a nontrivial solution, but it also will have infinitely many solutions. It is also possible, but not required, to have a nontrivial solution if $n = m$ and $n < m$.



Example 23:

Find the nontrivial solutions to the following homogeneous system of equations

$$\begin{aligned}2x + y - z &= 0 \\x + 2y - 2z &= 0\end{aligned}$$

Solution

Notice that this system has $m = 2$ equations and $n = 3$ variables, so $n > m$. Therefore by our previous discussion, we expect this system to have infinitely many solutions.

The process we use to find the solutions for a homogeneous system of equations is the same process we used in the previous section. First, we construct the augmented matrix, given by

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 1 & 2 & -2 & 0 \end{array} \right]$$

Then, we carry this matrix to its reduced row-echelon form, given below.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

The corresponding system of equations is

$$\begin{aligned} x &= 0 \\ y - z &= 0 \end{aligned}$$

Since z is not restrained by any equation, we know that this variable will become our parameter. Let $z = t$ where t is any number. Therefore, our solution has the form

$$\begin{aligned} x &= 0 \\ y &= z = t \\ z &= t \end{aligned}$$

Hence this system has infinitely many solutions, with one parameter t .

Suppose we were to write the solution to the previous example in another form. Specifically,

$$\begin{aligned} x &= 0 \\ y &= 0 + t \\ z &= 0 + t \end{aligned}$$

can be written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Remark 7.1. Notice that we have constructed a column from the constants in the solution (all equal to 0), as well as a column corresponding to the coefficients on t in each equation. While we will discuss this form of solution more in further chapters, for now consider the

column of coefficients of the parameter t . In this case, this is the column $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

There is a special name for this column, which is basic solution. The basic solutions of a system are columns constructed from the coefficients on parameters in the solution. We often denote basic solutions by X_1, X_2 etc., depending on how many solutions occur. Therefore,

Example above has the basic solution $X_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

We explore this further in the following example.

Consider the following homogeneous system of equations.